Harvard University

Physics 141 Pset 6

Due November 24, 2025, 11:59 PM

1. As you studied in class, hair cells turn movements of hair bundles into electrical activity. Hair bundles can be driven via random thermal forces from the surrounding fluid or by sound. Let $x(t)$ denote the hair-bundle displacement. In our model, there are three forces acting on hair bundles: (1) thermal fluctuations $\eta(t)$, (2) a spring-like restoring force $-\kappa x$, and (3) viscous drag force $-\gamma \frac{dx}{dt}$. Therefore, the dynamics of a hair bundle can be described via the following differential equation.

$$\gamma \frac{dx}{dt} + \kappa x = \eta(t)$$

We assume that the thermal force $\eta(t)$ is white, meaning

$$\langle \eta(t)\eta(t') \rangle = 2\gamma k_b T \delta(t - t')$$

The motion of a hair cell is referred to as Brownian motion in an elastic well. For this problem, define the Fourier transform in the Hz convention. In other words, for $x(t)$, define its Fourier transform $X(f)$ as

$$\mathcal{F}(x(t))(f) = X(f) = \int_{-\infty}^{\infty} dt\, x(t) e^{-2\pi i f t}$$

And define the inverse Fourier transform as

$$x(t) = \int_{-\infty}^{\infty} df\, X(f) e^{2\pi i f t}$$

(a) Show that $\mathcal{F}(\frac{dx}{dt}) = 2\pi i f X(f)$.

(b) By taking a Fourier transform of the differential equation $\gamma \frac{dx}{dt} + \kappa x = \eta(t)$, show that

$$X(f) = \frac{H(f)}{2\pi i \gamma f + \kappa}$$

Where $H(f)$ denotes the Fourier transform of $\eta(t)$.

As you saw in class, we can also define something called the two-sided power spectral density (PSD). Formally, the two-sided PSD for a signal $x(t)$ is defined as

$$S_{xx}^{(2)}(f) = \lim_{T\to\infty} \langle \frac{1}{T} X_T(f) X_T^*(f) \rangle$$

Where $X_T(f) = \int_{-T/2}^{T/2} dt\, x(t) e^{-2\pi i f t}$. The PSD is a measure of how the variance of x is distributed over frequencies. The PSD satisfies

$$\langle X_T(f) X_T^*(f') \rangle = T\delta(f - f') S_{xx}^{(2)}(f)$$

We can calculate the PSD using the Wiener-Khinchin Theorem, which states that the PSD equals to the Fourier transform of the autocorrelation:

$$S_{xx}^{(2)}(f) = \mathcal{F}(\langle x(t)x(t+\tau)\rangle)$$

When $x(t)$ is real, experiments often talk about the one-sided PSD instead of the two-sided. This one-sided PSD is defined for $f \geq 0$ and is simply $S_{xx}^{(1)} = 2S_{xx}^{(2)}$.

We can also define something called a cross-PSD between two signals $x(t)$ and $v(t)$ as

$$S_{vx}^{(2)}(f) = \lim_{T\to\infty} \langle \frac{1}{T} V_T(f) X_T^*(f) \rangle$$

Which satisfies $\langle V_T(f) X_T^*(f') \rangle = \delta(f - f') S_{vx}^{(2)}(f)$. The one-sided cross PSD is simply given by $S_{vx}^{(1)} = 2S_{vx}^{(2)}$.

(c) By combining your result from (b) and the Wiener-Khinchin Theorem, show that

$$S_{xx}^{(1)}(f) = \frac{4\gamma k_b T}{(2\pi f \gamma)^2 + \kappa^2}$$

(d) The result in (c) is called a Lorentzian. Explain in 2–3 sentences (as if to a biology student) what this spectrum says about the motion at low vs. high frequencies. We define $f_k = \frac{\kappa}{2\pi\gamma}$ as the "roll-off" frequency. Explain what this frequency means physically.

As you studied in class, Denk and Webb had performed experiments on the hair cells in frogs. They used a hair bundle's own Brownian motion as a built-in noisy input and asked: how much of the cell's voltage, denoted $v(t)$, is just the noisy input being transduced? Let $I(t)$ denote a causal impulse response of the cell, and $n(t)$ any added electrical noise. Treating the cell as linear and time-invariant, the voltage is a filtered version of displacement and is given by

$$v(t) = (I * x)(t) + n(t)$$

(e) For now, let's assume $n(t) = 0$. Suppose you drive the hair bundle with white noise $x(t)$ whose autocorrelation satisfies

$$\langle x(t)x(t+\tau)\rangle = \sigma_x^2\delta(\tau)$$

Show that the cross-correlation between the voltage you measure and $x(t)$ satisfies

$$\langle v(t)x(t+\tau)\rangle = \sigma_x^2 I(\tau)$$

This is a classic trick where for white driving noise, the cross-correlation between input and output equals the impulse response (up to a scale). We will now tackle the problem for the case where $x(t)$ is not white.

(f) Suppose now that $x(t)$ is not necessarily white noise and that electrical noise $n(t)$ is nonzero. Assume that $n(t)$ is not correlated with $x(t)$. Let $X(f)$ denote the Fourier transform of $x(t)$ and $V(f)$ the Fourier transform of $v(t)$. Also, let $X^*(f)$ denote the complex conjugate of $X(f)$. Show that

$$\frac{S_{vx}^{(2)}(f)}{S_{xx}^{(2)}(f)} = \hat{I}(f)$$

Where $\hat{I}(f)$ is the Fourier transform of the impulse response $I(t)$. Hint: it helps to apply a Fourier transform to both sides of the linear time-invariant model for voltage ($v(t) = (I * x)(t) + n(t)$).

(g) Prove the following inequality

$$|S_{vx}^{(2)}(f)|^2 \leq S_{vv}(f)S_{xx}(f)$$

There are a few ways to prove this. One way is to first let $y(t) = v(t) - \alpha x(t)$ for some complex number α. Show that $S_{yy}^{(2)}(f) = S_{vv}^{(2)}(f) - \alpha S_{xv}^{(2)}(f) - \alpha^* S_{vx}^{(2)}(f) + |\alpha|^2 S_{xx}^{(2)}(f)$. Choose α such that $S_{yy}^{(2)}(f)$ is minimized, which results in the desired inequality. Another option is to write the integral equations for the PSD and apply the Cauchy-Schwarz inequality directly (what you're proving is essentially a direct consequence of the Cauchy-Schwarz inequality).

Inspired by the fact that $|S_{vx}^{(2)}(f)|^2 \leq S_{vv}^{(2)}(f)S_{xx}^{(2)}(f)$, we define something called the coherence.

$$C(f) = \frac{|S_{vx}^{(2)}(f)|^2}{S_{vv}^{(2)}(f)S_{xx}^{(2)}(f)}$$

(h) Explain why $C(f)$ is between 0 and 1. Explain physically/intuitively what $C(f) \approx 1$ and $C(f) << 1$ mean.

We will now evaluate the formulas we just derived with physically realistic numbers. Suppose a frog hair bundle shows a root-mean-squared displacement of $\sqrt{\langle x^2 \rangle} = 2$ nm and a roll-off frequency of $f_k = 450$ Hz at room temperature.

(i) Using the equipartition theorem, estimate κ.

(j) Use the formula for f_k to estimate γ.

(k) Evaluate $S_{xx}^{(1)}(0)$ and $\sqrt{S_{xx}^{(1)}(0)}$ (the amplitude spectral density). Suppose you use a (shot-noise-limited) interferometry technique like in the Denk and Webb paper with a 1 pm/$\sqrt{Hz}$ sensitivity. Is this sensitive enough to resolve the thermal motion?

2. In class, you studied Hopf bifurcations. In this problem, you will explore a different type of bifurcation with interesting properties called the logistic map. We can model the growth of a population via a simple discrete update rule. Let x_n denote some measure of the number of animals in a population at timestep n. Our update rule is

$$x_{n+1} = f_r(x_n) = r x_n (1 - x_n)$$

Where $x_0 \in (0, 1)$ and $0 < r \leq 4$.

(a) Find all fixed points x^* of f_r and determine whether they are (linearly) stable. A fixed point is a point x^* that satisfies $x^* = f_r(x^*)$. A fixed point is stable when $|f_r'(x^*)| < 1$.

We will now generate a bifurcation diagram by simulating this system. In a programming language of your choice, do as follows.

- Take a grid of 1000 points of r-values between 2.5 and 4 ($2.5 \leq r \leq 4$).
- Set $x_0 = 0.5$.
- For each r, find x_n for 500 timesteps. Discard the first 300 timesteps. For the last 200 timesteps, plot x_n as a function of r.

(b) Describe the bifurcation diagram at various points (for example, r = 2.9, 3.2, 3.5, 3.83, 3.9).